

Active Directory VM Configuration in Azure

This step-by-step guide demonstrates how to configure and deploy a Windows Server VM in Microsoft Azure with networking, storage, and management options suitable for Active Directory.

Step 1: Configure Basic Settings

Set up the basic configuration of your VM, including the image, zone, and VM size. Choose Windows Server 2022 and select appropriate sizing (Standard_D2s_v3).

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Create a virtual machine

 Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.



Help me create a low cost VM

Help me create a VM optimized for high availability

Help me choose the right VM size for my workload

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

Availability zone  

Zone 1 

 You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type ⓘ

Trusted launch virtual machines

Configure security features

Image

Windows Server 2022 Datacenter: Azure Edition Hotpatch - x64 Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

☐ Arm64

 x64

 Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ

Size

Standard_D2s_v3 - 2 vcpus, 8 GiB memory (US\$145.27/month)

[See all sizes](#)

Enable Hibernation ⓘ

☐

i Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernate to enable this feature. [Learn more](#)

Step 2: Administrator Settings

Enter a username and secure password for your virtual machine administrator account. Enable RDP port (3389) to allow remote access.

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[See all sizes](#)

Enable Hibernation ⓘ

☐

ⓘ

Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#) ⓘ

Administrator account

Username * ⓘ

Tayvadi✓

Password *

*****✓

Confirm password *

*****✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☐ None

☒ Allow selected ports

Select inbound ports *

RDP (3389)▼

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Step 3: Disk Configuration

Select the type of OS disk (e.g., Premium SSD) and optionally configure encryption or attach additional data disks.

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Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size

OS disk type *

Delete with VM ☒

Key management

Enable Ultra Disk compatibility ☐

Data disks

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM
-----	------	------------	-----------	--------------	----------------

[Create and attach a new disk](#) [Attach an existing disk](#)

Step 4: Create a Virtual Network

Set up a virtual network and define address space and subnets for secure communication.

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Help me create a VM optimized for high availability

BasicsDisksNetworkingManagementMonitoringAdvancedTagsReview

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings, inbound and outbound connectivity with security group rules, or place behind an existing load balancer. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *

Create new

Public IP

None

Create new

NIC network security group

None

Basic

Advanced

Public inbound ports *

None

Allow selected ports

Select inbound ports *

RDP (3389)

This will allow all IP addresses to access your virtual machine. This is not recommended for testing. Use the Advanced controls in the Security group to create rules to limit inbound traffic to known IP addresses.

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Next : Management >

Review + create

Create virtual network

The Microsoft Azure Virtual Network service enables Azure resources to securely communicate with each other in a virtual network which is a logical isolation of the Azure cloud dedicated to your subscription. You can connect virtual networks to other virtual networks, or your on-premises network. [Learn more](#)

Name *

Address space

The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).

Address range *

Addresses

Overlap

10.0.0.0/16

10.0.0.0 - 10.0.255.255 (65536 addresses)

None

(0 Addresses)

None

Subnets

The subnet's address range in CIDR notation. It must be contained by the address space of the virtual network.

Subnet name

Address range

Addresses

default

10.0.0.0/24

10.0.0.0 - 10.0.0.255 (256 addresses)

(0 Addresses)

OK

Discard

Step 5: Networking Configuration

Assign the newly created virtual network and subnet to your VM and configure public IP and network security group (NSG).

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Create a virtual machine

Help me create a low cost VM

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Help me choose the right VM size for my workload

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Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
[Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *

(new) WindowsServer-VNet

Create new

Subnet *

(new) default (10.0.0.0/24)

Public IP

None

Create new

NIC network security group

None

Basic

Advanced

Public inbound ports *

None

Allow selected ports

Select inbound ports *

RDP (3389)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to

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[Give feedback](#)

Step 6: Management Options

Enable management tools like Microsoft Defender for Cloud and configure identity, auto-shutdown, and backup settings.

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Configure management options for your VM.

Microsoft Defender for Cloud

Microsoft Defender for Cloud provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

Enable basic plan for free

☒

This will apply to every VM in the selected subscription

Identity

Enable system assigned managed identity

☐

Microsoft Entra ID

Login with Microsoft Entra ID

☐

RBAC role assignment of Virtual Machine Administrator Login or Virtual Machine User Login is required when using Microsoft Entra ID login. [Learn more](#)

Auto-shutdown

Enable auto-shutdown

☐

Backup

Enable backup

☐

Step 7: Monitoring Settings

Choose monitoring options like boot diagnostics and OS guest diagnostics.

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Configure monitoring options for your VM.

Alerts

Enable recommended alert rules

Diagnostics

Boot diagnostics

- Enable with managed storage account (recommended)
- Enable with custom storage account
- Disable

Enable OS guest diagnostics

Health

Enable application health monitoring

Step 8: Advanced Configuration

Add custom scripts or extensions to automate your VM setup and configuration.

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Add additional configuration, agents, scripts or applications via virtual machine extensions or cloud-init.

Extensions

Extensions provide post-deployment configuration and automation.

Extensions ⓘ

Select an extension to install

VM applications

VM applications contain application files that are securely and reliably downloaded on your VM after deployment. In addition to the application files, an install and uninstall script are included in the application. You can easily add or remove applications on your VM after create. [Learn more](#)

Select a VM application to install

Custom data

Pass a script, configuration file, or other data into the virtual machine **while it is being provisioned**. The data will be saved on the VM in a known location. [Learn more about custom data for VMs](#)

Custom data

ⓘ

Your image must have a code to support consumption of custom data. If your

Step 9: Add Tags

Use tags to organize and categorize your resources efficiently. Tags are key-value pairs applied across Azure services.

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Tags are name/value pairs that enable you to categorize resources and view consolidated billing by applying the same tag to multiple resources and resource groups. [Learn more about tags](#)

Note that if you create tags and then change resource settings on other tabs, your tags will be automatically updated.

Name	Value	Resource
	:	14 selected

Step 10: Deployment Overview

Review deployment details and ensure all components (VM, NSG, VNet, etc.) are created successfully.

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

tayvadi.nzonza@ynov.c...
YNOV (YNOV.COM)

Home >

CreateVm-MicrosoftWindowsServer.WindowsServer-202-20250516134828 | Overview

Deployment

Search

Delete

Cancel

Redeploy

Download

Refresh

Overview

Inputs

Outputs

Template

Deployment is in progress

Deployment name: CreateVm-MicrosoftWindowsServer.WindowsS...
Subscription: Azure for Students
Resource group: ResourceGroup1

Start time: 16/05/2025, 13:59:47
Correlation ID: 0d67bdaf-7ec7-4c70-977c-cc06ea9eab83

Deployment details

Resource	Type	Status	Operation details
YNOV-WindowsServer	Microsoft.Compute/virtualMac...	Created	Operation details
ynov-windowsserver492_z1	Microsoft.Network/networkInt...	Created	Operation details
YNOV-WindowsServer-nsg	Microsoft.Network/networkSec...	OK	Operation details
WindowsServer-VNet	Microsoft.Network/virtualNetw...	OK	Operation details
YNOV-WindowsServer-ip	Microsoft.Network/publicIpAd...	OK	Operation details

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